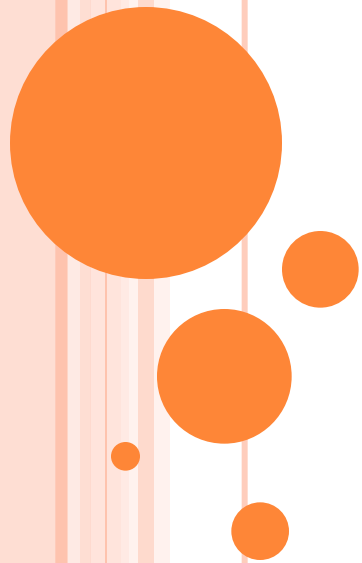


JOINT DISTRIBUTIONS



OUTLINE

- Joint Distributions (for two or more r.v.'s)
- Marginal Distributions
(computed from a joint distribution)
- Conditional Distributions (e.g. $P(Y = y | X = x)$)
- Independence for r.v.'s X and Y



JOINT DISTRIBUTIONS

- The statistical analysis for the relationship of two or more variables– Multivariate Analysis

Ex: Doctor may require height, weight, blood pressure, Sugar levels, etc.

- If it is about two variables of one experiment-- Bivariate Analysis

- height and weight of a person;
- temperature and rainfall of a day;
- two coordinates of a needle randomly dropped on a table;
- the number of 1s and the number of 6s in 10 rolls of a die.



JOINT PROBABILITY FUNCTION

- If X and Y are two random variables, the probability distribution that defines their simultaneous behavior is called a joint probability distribution.
- Let (X, Y) be the bivariate random variable defined on the sample space of an experiment. The *joint probability function* $f(x, y)$ is defined for each pair of numbers (x, y) by

$$f(x, y) = P(X = x \text{ and } Y = y)$$



Joint Probability Mass Function

Let X and Y be two discrete rv's defined on the sample space of an experiment. The *joint probability mass function* $p(x, y)$ is defined for each pair of numbers (x, y) by

$$p(x, y) = P(X = x \text{ and } Y = y)$$



PROPERTIES

$$1) 0 \leq P(x_i, y_j) \leq 1$$

$$2) \sum_i \sum_j P(x_i, y_j) = 1$$

$$3) P[x_2 \leq x \leq x_5, y_3 \leq y \leq y_4] = \sum_{x_2}^{x_5} \sum_{y_3}^{y_4} P(x, y)$$

$$4) F_{XY} = P[x \leq x_i, y \leq y_j] = \sum_{x_1}^{x_i} \sum_{y_1}^{y_j} P(x, y)$$


Marginal Probability Mass Functions

The *marginal probability mass functions* of X and Y , denoted $p_X(x)$ and $p_Y(y)$ are given by

$$p_X(x) = \sum_y p(x, y) \quad p_Y(y) = \sum_x p(x, y)$$



■ **Example.** We are interested in the effect of seat belt use on saving lives. If we consider the following random variables X and Y defined as follows:

- $X = 0$ if child survived
- $X = 1$ if child did not survive

- And $Y = 0$ if no belt
- $Y = 1$ if adult belt used
- $Y = 2$ if child seat used



- The following table represents the *joint probability distribution* of X and Y . In general we write $P(X = x, Y = y) = p(x, y)$ and call $p(x, y)$ the *joint probability function* of (X, Y) .

		X	
		0	1

Y	0	0.38	0.17 0.55
	1	0.14	0.02 0.16
	2	0.24	0.05 0.29

		0.76	0.24



- Probability that a child will both survive and be in a child seta when involved in an accident is:

$$P(X=0, Y=2) = 0.24$$

- Probability that a child will be in a child seat:

$$\begin{aligned} P(Y=2) &= P(X=0, Y=2) + P(X=1, Y=2) \\ &= 0.24+0.05= 0.29 \end{aligned}$$



Joint Probability Density Function

Let X and Y be continuous rv's. Then $f(x, y)$ is a *joint probability density function* for X and Y if for any two-dimensional set A

$$P[(X, Y) \in A] = \iint_A f(x, y) dx dy$$

If A is the two-dimensional rectangle
 $\{(x, y) : a \leq x \leq b, c \leq y \leq d\},$

$$P[a \leq x \leq b, c \leq y \leq d] = \int_a^b \int_c^d f(x, y) dy dx$$



Marginal Probability Density Functions

The *marginal probability density functions* of X and Y , denoted $f_X(x)$ and $f_Y(y)$, are given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{for } -\infty < x < \infty$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx \quad \text{for } -\infty < y < \infty$$



PROPERTIES

$$1) f_{XY}(x, y) \geq 0$$

$$2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$3) F_{XY} = P[x \leq x_i, y \leq y_j] = \int_{-\infty}^{x_i} \int_{-\infty}^{y_j} f(x, y) dx dy$$

$$4) f_{XY} = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$



(a) The joint probability function of two R.V's X & Y is given by

$$f(x, y) = \begin{cases} c(x^2 + 2y) & x = 0, 1, 2 \\ & y = 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases} \quad \text{find}$$

- i. The value of C
- ii. $P(x=2, y=3)$
- iii. $P(x \leq 1, y > 2)$
- iv. Marginal probability function of X & Y.



Independent Random Variables

Two random variables X and Y are said to be *independent* if for every pair of x and y values

$$p(x, y) = p_X(x) \cdot p_Y(y)$$

when X and Y are discrete or

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

when X and Y are continuous. If the conditions are not satisfied for all (x, y) then X and Y are *dependent*.



Conditional Probability Function

Let X and Y be two continuous rv's with joint pdf $f(x, y)$ and marginal X pdf $f_X(x)$. Then for any X value x for which $f_X(x) > 0$, *the conditional probability density function of Y given that $X = x$ is*

$$f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)} \quad -\infty < y < \infty$$

If X and Y are discrete, replacing pdf's by pmf's gives the *conditional probability mass function of Y when $X = x$.*



Let X and Y denote the proportion of two different chemicals in a sample mixture of chemicals used as an insecticide. Suppose X and Y have joint probability density given by:

$$f(x, y) = \begin{cases} 2, & 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq x + y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

(Note that $X + Y$ must be at most unity since the random variables denote proportions within the same sample).

- 1) Find the marginal density functions for X and Y .
- 2) Are X and Y independent?
- 3) Find $P(X > 1/2 \mid Y = 1/4)$.



- Marginal density functions are

$$f_1(x) = \begin{cases} \int_0^{1-x} 2dy = 2(1-x), & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad f_2(y) = \begin{cases} \int_0^{1-y} 2dx = 2(1-y), & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- 2) $f_1(x) f_2(y) = 2(1-x) * 2(1-y) \neq 2 = f(x,y)$, for $0 \leq x \leq 1-y$.
- Therefore X and Y are not independent.

- 3)

$$P\left(X > \frac{1}{2} \mid Y = \frac{1}{4}\right) = \int_{1/2}^1 f(x \mid y = \frac{1}{4}) dx = \int_{1/2}^1 \frac{f(x, y = \frac{1}{4})}{f(y = \frac{1}{4})} dx = \int_{1/2}^1 \frac{2}{2(1 - \frac{1}{4})} dx = \frac{2}{3}$$



Example. Let the joint density of two random variables x and y be given by

$$f(x, y) = \begin{cases} \frac{1}{6}(x + 4y) & 0 < x < 2, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- 1) Find the marginal density functions for X and Y .
- 2) Are X and Y independent?
- 3) Find $P(X < 1 \mid Y = 1/2)$.



(a) Joint probabilities of two random variables X and Y are given in table 3a

$\begin{array}{c} X \\ Y \end{array}$	1	2	3
1	$1/7$	$3/28$	$1/14$
2	$1/7$	$3/28$	$1/14$
3	$1/14$	$2/14$	$1/7$

Table 3a

Find

- $P(X \leq 1.5)$
- XY is even
- Y is odd given that X is even.



A1, A2 and A3 are three mutually exclusive and exhaustive sets of events associated with a random experiment E1. Events B1, B2 and B3 are mutually exclusive and exhaustive sets of events associated with a random experiment E2. The joint Probabilities of occurrence of these events and some marginal probabilities are listed in the table given below:

	B1	B2	B3
A1	$3/36$	*	$5/36$
A2	$5/36$	$4/36$	$5/36$
A3	*	$6/36$	*
P(Bj)	$12/36$	$14/36$	*

- Find the missing probabilities (*) in the table.
- Find $P(B3|A1)$ and $P(A1|B3)$
- Are events A1 and B1 statistically independent?

[4+4+2]

